

SUMANTH R

Chennai, Tamil Nadu
B.Tech. in Computer Science and Engineering (AIML)
SRM Institute of Science and Technology

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Education

Degree	Institute	CGPA	Duration
B.Tech., Computer Science and Engineering (AIML)	SRM Institute of Science and Technology, Chennai, Tamil Nadu	8.98 / 10	*2023 – 2027*

Experience

- **Works Applications India Pvt. Ltd.** *On-site*
  Software Engineering Intern Dec 2025 – Jan 2026
  - Analyzed application data using MySQL and optimized SQL queries, reducing response time by **25%**.
  - Developed **5+ REST APIs** supporting **5K+ simulated requests/day** using Java, Spring Boot, and MySQL.
  - Integrated Explainable AI insights into the frontend to improve model interpretability and user decision-making.

- **All India Council for Technical Education (AICTE)** *Remote*
  Google AI/ML Virtual Intern Jul 2024 – Sep 2024
  - Analyzed **5+ datasets** through preprocessing, feature engineering, and exploratory data analysis.
  - Built and evaluated **8+ ML models**, achieving up to **92% accuracy** across supervised and unsupervised tasks.
  - Applied hyperparameter tuning and model evaluation techniques, improving performance by up to **25%**.

Projects

- **E-Commerce Sales & Customer Analytics Dashboard** *Jul 2026*
  Technologies: Power BI, SQL, Excel, Python, Pandas
  - Analyzed sales and customer data using SQL and Python to identify revenue trends and purchasing patterns.
  - Built a Power BI dashboard tracking revenue, profit, AOV, retention, and regional performance.
  - Segmented customers and products to identify high-value groups and actionable business insights.

- **Retail Sales Analysis using SQL** *Jul 2026*
  Technologies: SQL, MySQL, Excel
  - Analyzed sales data using joins, CTEs, subqueries, aggregations, and window functions.
  - Evaluated product, customer, and regional performance using revenue, growth, and profitability KPIs.
  - Created Excel reports and visualizations to communicate trends and business insights.

- **Customer Churn Analysis** *Jul 2026*
  Technologies: Python, Pandas, NumPy, Matplotlib, Scikit-learn
  - Performed data cleaning and EDA to identify key factors influencing customer churn.
  - Analyzed customer segments, usage patterns, and churn trends using Python and statistical methods.
  - Built predictive models and derived actionable insights to support customer retention.

- **Marine Sentinel: Oil Spill Analytics System** *Mar 2026*
  Technologies: Python, Streamlit, OpenCV, PyTorch, YOLOv6
  - Analyzed **5,300+ UAV frames** to detect and quantify oil spill patterns.
  - Achieved **97% mAP50** and **86% mAP50–95**, outperforming 4+ baselines.
  - Built a Streamlit dashboard for spill detection, coverage, and severity analysis.

Publications

- **Deep Learning Based Brain Tumor Classification using Orange Data Analytics** *2026*
  IEEE International Conference (SCEECS 2026) View Paper
  - Designed an end-to-end deep learning pipeline on a **16,000+ MRI image dataset** aggregated from Kaggle, BRISC, and FigShare.
  - Achieved **96%+ classification accuracy** and **AUC up to 0.997** using Neural Networks with stratified 10-fold cross-validation.
  - Reduced feature dimensionality by **75%** using PCA while retaining **95% variance**, improving training efficiency.
  - Improved model performance by **+7% over baseline methods** and enhanced MCC by **+8%** using data augmentation techniques.

Achievements

- Selected as a Software Engineering Fellow at Headstarter AI, collaborating on real-world software projects.
- Secured 3rd Place in CodeMavens 2025 hackathon organized by ACM-W SRM Student Chapter.
- Built strong connections through content creation, leading to mentorship and collaboration opportunities.

Technical Skills

- **Data Analytics**: SQL, Python, MS Excel, Pandas, NumPy, Exploratory Data Analysis (EDA), Data Cleaning, Data Validation
- **Business Intelligence**: Power BI, Tableau, Dashboard Development, KPI Analysis, Data Visualization, Business Reporting
- **SQL & Databases**: MySQL, SQL Queries, Joins, Subqueries, CTEs, Window Functions, Aggregations, Query Optimization
- **Excel**: Pivot Tables, XLOOKUP, VLOOKUP, INDEX-MATCH, Conditional Formatting, Charts, Data Cleaning
- **Analytics & ML**: Statistical Analysis, Descriptive Statistics, Scikit-learn, Machine Learning, Feature Engineering, Model Evaluation
- **Developer Tools**: Jupyter Notebook, Git, GitHub, VS Code, Linux
- **Core Concepts**: DBMS, Data Structures & Algorithms, OOP, Problem Solving